KPR Institute of Engineering and Technology

Department of CSE (Artificial Intelligence and Machine Learning)

Project Report

on

U21AM502

INTERNET AND WEB PROGRAMMING

Submitted in partial fulfillment of the requirement for the award of the Degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

(Specialization in Artificial Intelligence and Machine Learning)

by

BALA GANESH M (24AM007)

Department of CSE (Artificial Intelligence and Machine Learning)

NOV 2026-2027

CANDIDATE'S DECLARATION

I hereby declare that the client-based project report titled "**THE CAKE HOME – CAKE BUSINESS WEBSITE**" submitted to the Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning), KPR Institute of Engineering and Technology, in partial fulfillment of the requirements for the award of the Degree of Bachelor of Engineering, is a bonafide record of authentic software engineering work carried out by me under the supervision of **Dr. KARTHICK PANNEERSELVAM**, Associate Professor, Department of CSE (AIML).

I further confirm that the client consultation records, design, front-end implementation, system architecture, testing, deployment, revenue generation details and deliverables documented herein represent a genuine technical engagement undertaken for The Cake Home, a cake business located in Hosur, Tamil Nadu. The production website is deployed and publicly accessible at https://surya2735151.github.io/the-cake-home/. Where any administrative or third-party operation is subject to client verification, it is explicitly noted as such rather than represented as completed. The content of this report has not been submitted previously for the award of any other degree, diploma, fellowship, or academic qualification.

Student Candidate Name & Register Number	Academic Signature & Submission Date
1. BALA GANESH M (Roll No: 24AM007) Department of CSE (AIML), KPRIET	Signature: _____ Date: 02 November 2026

Faculty Guide Name & Designation	Faculty Guide Signature & Date
Dr. KARTHICK PANNEERSELVAM Associate Professor Department of CSE (AIML), KPRIET	Signature: _____ Date: 02 November 2026

CLIENT APPRECIATION STATEMENT

I, **Mr. Akilan A**, Proprietor & Baker of **The Cake Home**, Hosur, Tamil Nadu, hereby state that student developer **BALA GANESH M (Roll No: 24AM007)** from KPR Institute of Engineering and Technology was engaged to design and develop the official website of our cake business, deployed at https://surya2735151.github.io/the-cake-home/.

The student developer held requirement discussions with us from August 2026, studied how we receive cake orders through WhatsApp, and delivered a responsive website featuring our cake collections and categories, a customized cake enquiry form that prepares a WhatsApp message, a baking classes section, and links to our Instagram and YouTube channels. The project was executed with professional diligence, adhered to our brand style, and the website was handed over to us for our own use.

The agreed commercial remuneration of ₹1,000 (Rupees Two Thousand only) for the website development work has been paid in full and settled on 30 October 2026, as recorded in the invoice and payment details of this report. I issue this statement as my genuine patronage and endorsement of this client-based project.

Client Representative Details	Enterprise Verification & Seal
Mr. Akilan A Proprietor & Baker The Cake Home, Hosur, Tamil Nadu Phone / WhatsApp: +91 95978 33830 Instagram: @the__cake_home Authorized Signature: _____ Date: 30 October 2026	Official Business Seal / Stamp

ACKNOWLEDGEMENT

I take immense pride and honor in presenting this comprehensive academic project report titled "**THE CAKE HOME – CAKE BUSINESS WEBSITE**", completed as part of the requirements for the course U21AM502 – INTERNET AND WEB PROGRAMMING during the academic year 2026–2027.

First and foremost, I express my sincere thanks to our Principal, **Dr. Akila Muthuramalingam**, and the management of KPR Institute of Engineering and Technology for providing the institutional support, resources, and academic environment necessary to enrich my engineering training.

I convey my deep gratitude to **Dr. Pandiya Rajan G**, Professor & Head of the Department of Computer Science and Engineering (AIML), for providing the laboratory infrastructure, computing facilities, and academic guidance that fostered my practical innovation.

I express my profound gratitude to my esteemed Faculty Guide, **Dr. Karthick Panneerselvam**, Associate Professor in the Department of Computer Science and Engineering (AIML), for his invaluable technical guidance, insightful design reviews, and continuous encouragement throughout this project.

Finally, I extend my heartfelt appreciation to my client, **Mr. Akilan A**, Proprietor of The Cake Home, Hosur, for his generous cooperation, patient domain discussions, and for providing the business information and cake ideas essential to completing this website.

ABSTRACT

The Cake Home Website is a modern, responsive, cloud-hosted web application engineered to give a Hosur-based cake business a professional digital presence and an organised channel for customized cake enquiries. Prior to this intervention, The Cake Home (Hosur, Tamil Nadu), which makes customized, birthday, anniversary, celebration and theme-based cakes, depended on word-of-mouth referrals and scattered social media posts. It had no single web address presenting its work, and cake requirements reached the baker as incomplete, unstructured messages that required repeated clarification.

To overcome these challenges, this project delivers an enquiry-based business website built with HTML5, CSS3 and JavaScript and published on GitHub Pages. The solution features a branded homepage, a cake collection with four categories, a customized cake enquiry form with required-field validation, WhatsApp integration (navigation button, floating icon and a pre-filled message generated from the form), a baking classes section, and direct links to the business's Instagram and YouTube channels. The website deliberately includes no online payment gateway, checkout or automated order processing; price discussion, order confirmation and payment remain directly between the customer and the business.

The site is hosted free on GitHub Pages with automatic HTTPS, so no domain or hosting expenditure was incurred. The commercial value of ₹1,000 was received in full on 30 October 2026, achieving a 100% net commercial realization, from which a charitable contribution of ₹1,000 was made to KPR Charities on 02 November 2026 as an allocation of net earnings.

Keywords: Client-Based Web Project, Responsive Website, Enquiry Form, WhatsApp Click-to-Chat Integration, GitHub Pages, Static Web Architecture, Revenue Model, Small Business Digital Presence.

TABLE OF CONTENTS

Section / ID	Academic Report Chapter Title & Description	Page
Cover	Cover Page (Institute, Degree, Candidate & Department Details)	1
Declaration	Candidate's Declaration (Student & Faculty Guide Signature Matrix)	2
Statement	Client Appreciation Statement (Mr. Akilan A, The Cake Home)	3
Ack	Acknowledgement (Faculty Guide Dr. Karthick Panneerselvam, Mentors & HOD)	4
Abstract	Executive Summary, Problem Context & Technical Highlights	5
Index	Table of Contents and Academic Deliverable Mapping	6
1	Project Abstract & Executive Summary	7
2	Client Organization Profile & Problem Definition	8
3	Requirement Analysis & Software Specification	9
4	Project Proposal, Work Breakdown Structure & Quotation	11
5	System Architecture, Design Language & Visual Hierarchy	13
6	Technical Implementation Deep-Dive	16
7	Verification, Quality Assurance & Testing Suite	19
8	Deployment, Web Operations & Low-Cost Infrastructure	20
9	Client Interaction, Meeting Records & Change Management	21
10	Revenue Generation Model & Payment Details	22
11	Client Verification & Authenticity Audit	26
12	Project Delivery, Training & Acceptance	26
13	Outcomes, Learning Scope, Future Roadmap & Conclusion	27
14	Institutional Evaluation Rubric, Checklist & Approvals	28
15	Comprehensive Annexures & Academic References	30
Annexure C	Invoice Details, Commercial Quotation, Testimonial & Payment Receipt	32
Annexure D	Academic Citations, Technical Frameworks & Platform References	35

1. PROJECT ABSTRACT & EXECUTIVE SUMMARY

1.1 Executive Summary

The Cake Home Website project addresses the operational bottleneck of enquiry handling, cake discovery and customer communication for The Cake Home, a home-scale cake business in Hosur, Tamil Nadu that specialises in customized, birthday, anniversary, celebration and theme-based cakes. Traditional operations relied on word-of-mouth referrals, social media posts and free-form messages, which created incomplete enquiries and repeated clarification. The student-developed responsive website establishes an elegant, professional and always-available digital presence deployed at https://surya2735151.github.io/the-cake-home/.

On the customer-facing side, the website delivers a branded homepage, an organised cake collection with four categories, a customized cake enquiry form with required-field validation, a baking classes section, and links to the business's Instagram and YouTube channels. The enquiry form places the entered details into a pre-filled WhatsApp message that the customer sends directly to the baker, using the communication channel the business already relies on. The website intentionally has no checkout or payment gateway; order confirmation, pricing and payment remain a direct conversation between customer and business.

1.2 Commercial Revenue Realization

In strict compliance with institutional requirements, this client project realized an agreed commercial value of ₹1,000 (Rupees Two Thousand only), received in full on 30 October 2026 through electronic payment (UPI). By leveraging free GitHub Pages hosting and open-source web technologies, 100% net commercial realization was attained. A charitable contribution of ₹1,000 was subsequently made to KPR Charities on 02 November 2026.

1.3 Technical Highlights Summary

Table 1.1: Core Technical & Operational Architecture Profile

Technical Dimension	Production Implementation Specification
Front-End Technologies	HTML5 semantic structure, CSS3 styling and vanilla JavaScript behaviour; static, single-page layout
Design System & Styling	Warm cream, cocoa-brown and rose palette with elegant serif headings and generous spacing
Data Tier	No database by design; the enquiry is placed directly into a WhatsApp message and no submission is stored
Resilience Strategy	Static hosting with no server runtime; Git version control allows instant rollback of any change
Hosting & Operations	GitHub Pages global CDN with automatic HTTPS and zero monthly hosting cost

Technical Dimension	Production Implementation Specification
External Integrations	WhatsApp Click-to-Chat (wa.me) links, Instagram profile link, YouTube channel link
Responsiveness	Adaptive layouts for desktop, tablet and mobile screens with a viewport-aware design
Production URL	https://surya2735151.github.io/the-cake-home/

2. CLIENT ORGANIZATION PROFILE & PROBLEM DEFINITION

2.1 Client Identity & Business Overview

The Cake Home (proprietor Mr. Akilan A) is a cake business located in Hosur, Tamil Nadu. It makes customized cakes, birthday cakes, anniversary cakes, celebration cakes and theme-based cakes, and also presents baking classes as a further line of engagement. The business serves families, friends and individuals planning celebrations in and around Hosur, and receives its orders and enquiries mainly through WhatsApp, with Instagram and YouTube used for visual marketing.

2.2 Baseline Operational Bottlenecks

- **Unstructured Enquiries:** Cake requests arrived as casual messages such as "How much for a cake?", without date, weight, theme or budget, forcing the baker to ask the same questions repeatedly.
- **No Single Professional Web Address:** The business's work was scattered across social media posts, so prospective customers had no organised place to explore cake styles or judge the brand.
- **Limited Discoverability:** Customers who heard about the business by word of mouth had no link to share that presented its range, location and contact details in one place.
- **Fragmented Channels:** Instagram, YouTube and WhatsApp operated independently, with no common destination connecting visual inspiration to a structured enquiry.

2.3 Mandatory Client Business Identification Matrix

Table 2.1: Client Enterprise Baseline & Identification Matrix (Mandatory Syllabus Format Table 1)

Identification Field	Client Enterprise Profile
Enterprise Name	The Cake Home
Proprietor & Baker	Mr. Akilan A
Business Location	Hosur, Tamil Nadu, India
Business Sector	Home-scale bakery: customized, birthday, anniversary, celebration and theme-based cakes; baking classes
Target Audience	Families, birthday and anniversary hosts, friends planning gatherings, aspiring home bakers

Identification Field	Client Enterprise Profile
Contact	WhatsApp / Phone: +91 95978 33830
Social Channels	Instagram: @the__cake_home; YouTube: @thecakehome
GST Status	Not applicable to this engagement (student development fee below registration threshold; no GST charged)
Commercial Valuation	₹1,000 (Subsidized curriculum client engagement, Settled 30 October 2026)
Core Digital Need	Professional web presence, structured cake enquiry form, WhatsApp bridge, social media integration

3. REQUIREMENT ANALYSIS & SOFTWARE SPECIFICATION

3.1 Problem Definition & Stakeholder Personas

The requirement engineering phase identified two primary stakeholders: **Customers** (seeking inspiration, a trustworthy impression and a frictionless way to ask for a cake) and the **Baker / Proprietor** (needing complete, well-organised requirements in the first message and no additional software to learn or maintain).

3.2 Functional Requirements Specification (FRS)

Table 3.1: Detailed Functional Requirements Specification (FRS)

Req ID	Module	Functional Requirement Description	Priority
FR-01	Catalog	Present the cake collection grouped into Birthday & theme, Chocolate, Anniversary & celebration and Custom creations	Critical
FR-02	Enquiry	Provide a customized cake enquiry form capturing name, contact, category, event date, theme, flavour, weight, reference image URL and budget	Critical
FR-03	Validation	Enforce required fields (name, contact number, cake category, event date, theme or design idea) before the enquiry can proceed	High
FR-04	Notification	Auto-generate a URL-encoded, pre-filled WhatsApp message from the form details addressed to the business	Critical
FR-05	WhatsApp CTAs	Offer "Order on WhatsApp" in the navigation bar, a floating WhatsApp icon and contextual "Ask us on WhatsApp" links	High
FR-06	Social Links	Link to the business Instagram account and YouTube channel from the contact section and footer	Medium
FR-07	Classes	Present Baking Classes ideas with per-class WhatsApp enquiry links	Medium
FR-08	Responsiveness	Render correctly on desktop, tablet and mobile with a navigation bar and skip-to-content link	High

3.3 Detailed Customer Use Case Narratives

Table 3.2: Formal Customer & Baker Use Case Narratives

Use Case ID	Actor	Description	Pre-conditions	Post-conditions
UC-01	Customer	Browse cake collection and categories	Website open	Customer views cake styles and ideas
UC-02	Customer	View about, classes and contact details	Website open	Business information and classes displayed
UC-03	Customer	Submit customized cake enquiry	Required fields completed	Pre-filled WhatsApp message opened
UC-04	Customer	Generate pre-filled WhatsApp message	Valid enquiry submitted	Encoded message ready in WhatsApp
UC-05	Customer	Open Instagram / YouTube channels	Website open	Social channel opens for more cake ideas
UC-06	Customer	Chat directly on WhatsApp	Website open	WhatsApp chat with the business opens
UC-07	Baker	Discuss price and design; confirm order	Enquiry received on WhatsApp	Order agreed directly with customer

The customer journey supported by the website proceeds through eight stages, set out in Table 3.3 with the website element involved at each stage and the outcome for the business.

Table 3.3: Customer Journey Supported by The Cake Home Website

Stage	Customer Action	Website Element Involved	Outcome for the Business
1	Visits The Cake Home website	Homepage hero section, brand identity and navigation bar	First professional impression of the brand
2	Explores the homepage	"Explore our cakes" call-to-action, welcome and brand-note sections	Visitor engagement with the business story
3	Browses cake designs	"Our Cakes" section and category presentation	Cake styles shown to a prospective customer
4	Selects a cake category	Cake category field in the enquiry form	Interest narrowed to a specific style
5	Submits enquiry or contacts on WhatsApp	Cake Enquiry form; "Order on WhatsApp" button; floating icon	Structured enquiry delivered to the business
6	Business receives requirements	Pre-filled WhatsApp message with the form details	Requirements received in one organised message
7	Business discusses price, design, availability	Takes place on WhatsApp, outside the website	Opportunity to quote and advise
8	Customer and business finalise the order	Takes place directly, outside the website	Possible conversion of the enquiry into an order

3.4 System Use Case Diagram

The use case diagram illustrates functional interactions between customers and the baker. The application boundary encapsulates catalogue browsing, enquiry validation, message generation and WhatsApp hand-off.

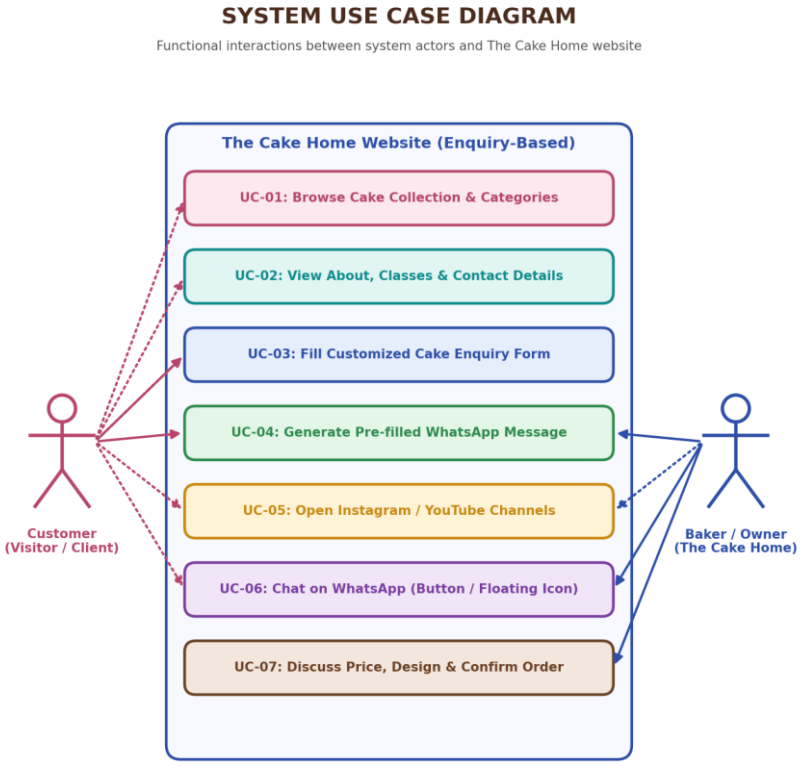


Figure 3.1: System Use Case Diagram of The Cake Home Website

3.5 Non-Functional Technical Requirements

- **Performance & Speed:** Fast first paint through a lightweight static site served from the GitHub Pages CDN.
- **Visual Aesthetics & Warmth:** A premium, celebratory character expressed through a cocoa-and-rose palette, elegant serif headings and the tagline "Made for your moments".
- **Privacy:** No customer data is stored; the form only places the entered details into a WhatsApp message that the customer chooses to send.
- **Accessibility & Usability:** Responsive viewport, semantic sections, a skip-to-content link and large tap targets for mobile users.
- **Maintainability:** A simple static code base under Git version control that the business can update without server administration.

4. PROJECT PROPOSAL, WORK BREAKDOWN STRUCTURE & QUOTATION

4.1 Scope Baseline & Explicit Boundaries

Table 4.1: Project Scope Baseline & Boundary Matrix

In Scope (Committed Commercial Deliverables)	Out of Scope (Explicit Boundaries)
Custom responsive storefront website with navigation bar	Online payment gateway integration (Razorpay / Stripe)
Cake collection and category presentation	Shopping cart, checkout or automated order processing
Customized cake enquiry form with required-field validation	Database, user accounts or admin dashboard
WhatsApp integration: button, floating icon, pre-filled message	Paid SMS / marketing automation gateways
Baking classes section, Instagram and YouTube links	Native mobile app distribution (App Store / Google Play)
Free GitHub Pages deployment with HTTPS and handover	Custom domain purchase and recurring maintenance

4.2 Work Breakdown Structure (WBS) & Milestones Schedule

The project ran from **10 August 2026 to 29 October 2026**, including client review intervals and final sign-off.

Table 4.2: Work Breakdown Structure (WBS) & Milestone Execution Timeline

Phase	Milestone Activity	Duration	Deliverables	Status
Phase 1	Client Consultation & Requirements	10 Days	Agreed scope, content list and problem matrix	Completed
Phase 2	UI/UX Prototyping & Design System	12 Days	Colour palette, typography and layout blueprint	Completed
Phase 3	Front-End Development & Responsiveness	18 Days	Homepage, navigation, cake gallery, classes section	Completed
Phase 4	Enquiry Form & WhatsApp Integration	15 Days	Validated form, wa.me message builder, social links	Completed
Phase 5	Testing, Quality Assurance & UAT	10 Days	Cross-browser and device checks, client UAT	Completed
Phase 6	GitHub Pages Deployment & Handover	05 Days	Production release and client sign-off	Completed

4.3 Commercial Quotation Baseline

The commercial quotation was developed on curriculum-mandated student development guidelines, delivering a complete professional website for The Cake Home at a highly subsidized commercial rate.

Table 4.3: Commercial Quotation & Billable Value Breakdown

Item	Service Description	Commercial Quoted Value	Settlement Status
1	Requirement Analysis and Project Planning	₹150	Paid & Settled
2	UI/UX Design and Website Layout	₹125	Paid & Settled
3	Homepage and Navigation Development	₹125	Paid & Settled
4	Cake Gallery and Category Integration	₹125	Paid & Settled
5	Customized Cake Enquiry Form	₹125	Paid & Settled
6	WhatsApp Integration	₹100	Paid & Settled
7	Responsive Website Development	₹100	Paid & Settled
8	Testing and Quality Assurance	₹75	Paid & Settled
9	Website Deployment and Final Handover	₹75	Paid & Settled
10	Domain & Hosting (GitHub Pages free subdomain)	₹0 (Free Tier)	Completed
TOTAL	**Total Agreed Client Project Value**	**₹1,000**	**100% Settled**

Table 4.4: Technical Deliverables Handover Matrix

Deliverable	Technical Scope & Verification	Handover Status
Live Website	Responsive static website deployed on GitHub Pages with HTTPS	Verified Publicly Active
Enquiry Form & WhatsApp Bridge	Required-field validation and pre-filled wa.me message	Verified Functionally Active
Social Integration	Instagram and YouTube links on contact section and footer	Verified Live
Source Code Repository	Complete Git repository with commit history	Verified Delivered to Client

5. SYSTEM ARCHITECTURE, DESIGN LANGUAGE & VISUAL HIERARCHY

5.1 High-Level Front-End System Architecture

The platform employs a lean, static, cloud-hosted architecture. The website is delivered as pre-built HTML, CSS and JavaScript files by GitHub Pages and rendered entirely in the visitor's browser. There is no application server and no database: the enquiry is composed in the browser and handed to WhatsApp through a Click-to-Chat link.

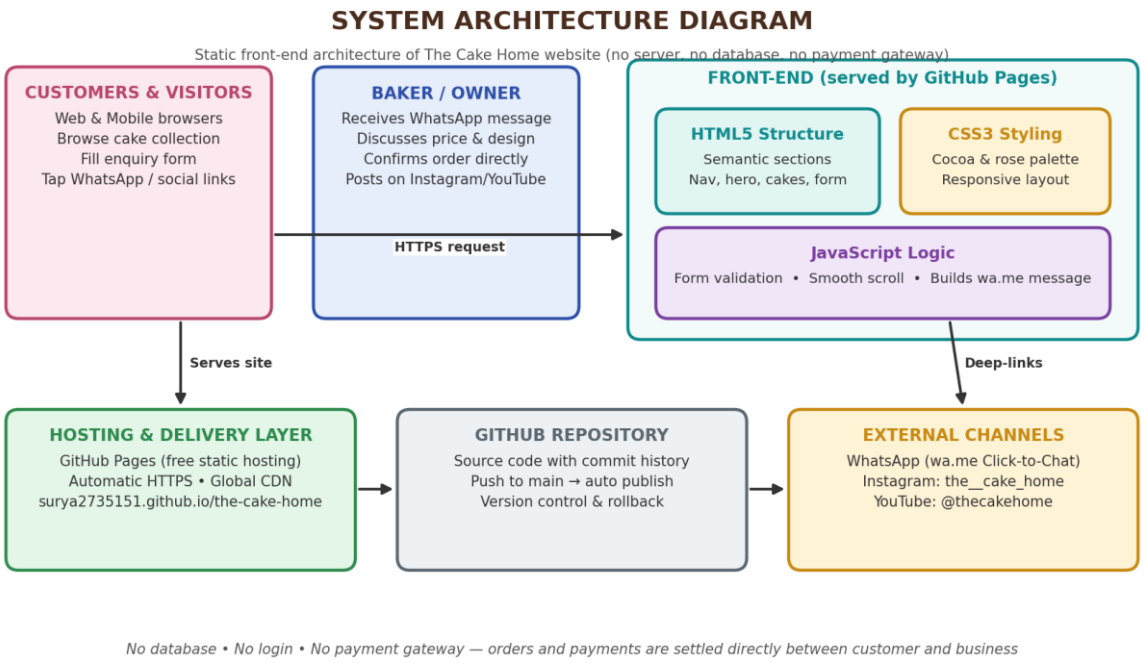


Figure 5.1: High-Level System Architecture Diagram

Table 5.1: System Architectural Modules & Responsibilities

Module	Technology Stack	Primary Architectural Responsibility
Presentation	HTML5, CSS3 (responsive layouts)	Delivers the branded, responsive user interface and cake presentation
Application Logic	Vanilla JavaScript in the browser	Validates required form fields, handles smooth scrolling and builds the encoded WhatsApp message
Hosting	GitHub Pages with HTTPS and CDN	Serves static files globally at no cost and publishes on every push
Notification	WhatsApp Click-to-Chat (wa.me)	Opens a chat with the structured enquiry pre-filled for the customer to send
Social	Instagram and YouTube links	Connects the website with the business's visual marketing channels

5.2 Data Flow & Logical Data Model

The Level-1 Data Flow Diagram (DFD) traces the flow of information from content browsing, through form validation and message construction, to the WhatsApp hand-off and direct negotiation with the baker.

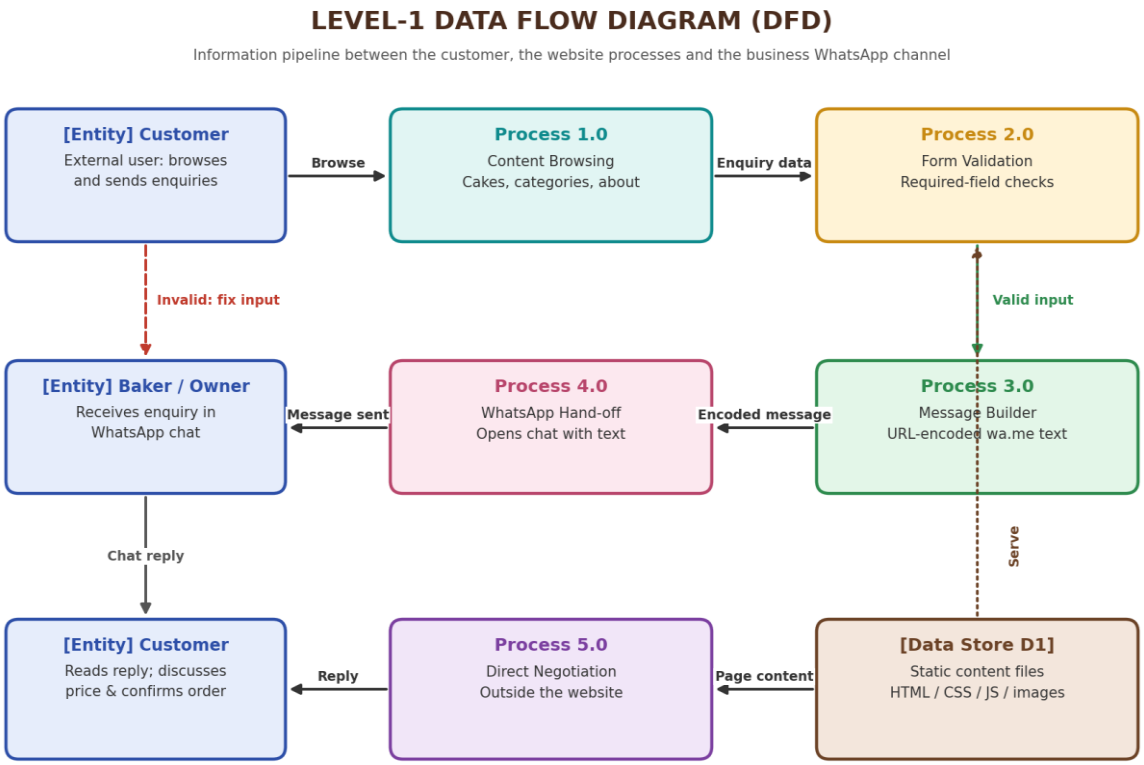


Figure 5.2: Level-1 Data Flow Diagram (DFD) of Enquiry & Notification Pipeline

5.3 Logical Content Model & Enquiry Data Structure

Because the website is intentionally database-free, its logical model describes the content entities rendered on the page and the structure of the enquiry that is transmitted through WhatsApp.

Table 5.2: Core Content Entities & Enquiry Data Specification

Entity	Identifier	Key Attributes	Storage
Cake Category	Section index (01–04)	title, tagline, description, image, link to enquiry form	Static HTML
Enquiry (Required)	Form field name	name, contact number, cake category, event date, theme or design idea	Browser memory only
Enquiry (Optional)	Form field name	flavour, approximate weight, reference image URL, budget range	Browser memory only
WhatsApp Message	wa.me/919597833830	URL-encoded text body composed from the enquiry fields	Not stored
Class Idea	Workshop index (01–03)	title, description, image, pre-filled WhatsApp enquiry link	Static HTML

5.4 Information Architecture & Visual Design Tokens

The structural blueprint below illustrates the hierarchical navigation order from the homepage down to the cake collection, enquiry form, brand sections and baking classes.

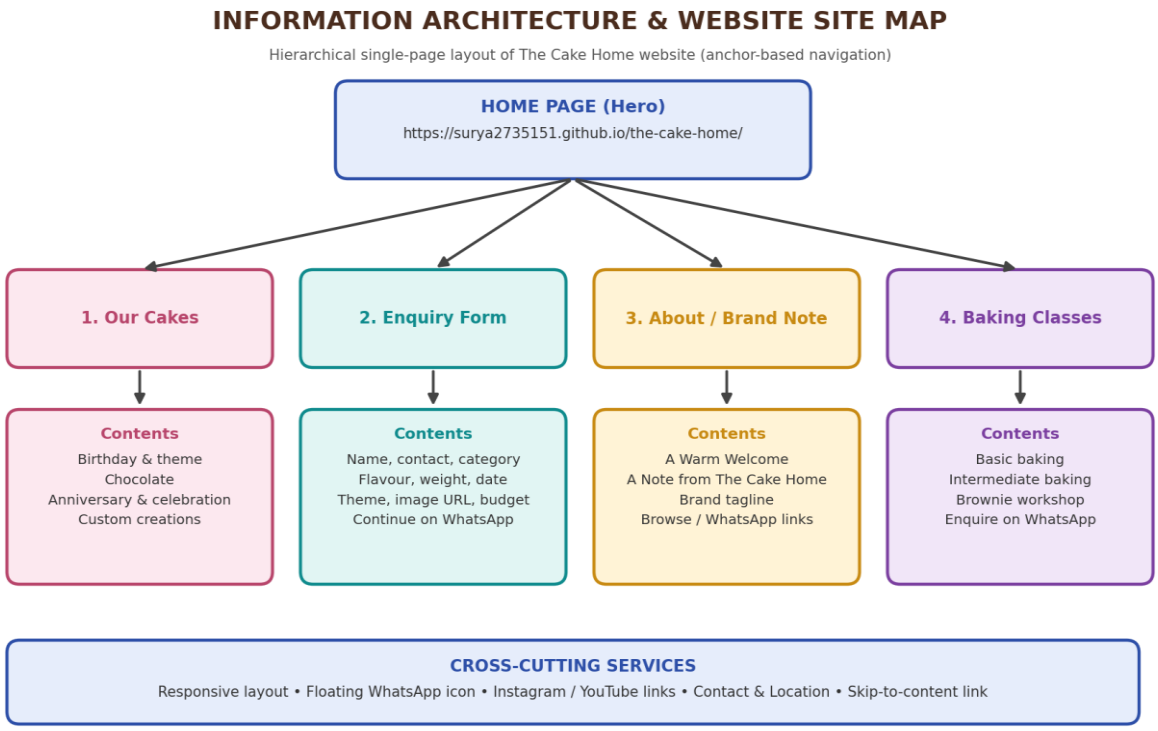


Figure 5.4: Information Architecture & Website Navigation Site Map

Table 5.3: Visual Design System, Typography & Colour Token Specification

Token Name	Colour / Spec	Design Role & Hierarchy
Canvas Background	Warm cream (#FFF9F4)	Soft page background and browser theme colour for a gentle, bakery-like feel
Cocoa Brown	Deep chocolate tone	Primary headings, body emphasis and brand identity text
Rose Accent	Soft rose tone	Calls to action, hover highlights and decorative details
Heading Typography	Elegant serif, italic accents	Celebratory headings such as "Every celebration deserves a special cake."
Body Typography	Clean sans-serif	High legibility for descriptions, form labels and mobile reading

6. TECHNICAL IMPLEMENTATION DEEP-DIVE

6.1 Core Technologies & Repository Architecture

The code base follows a clean separation of concerns: semantic HTML for structure, CSS for presentation and JavaScript for behaviour. The complete source is maintained in a Git repository and published through GitHub Pages at https://surya2735151.github.io/the-cake-home/.

Table 6.1: Repository Structure & Architectural Module Allocation

Component	Architectural Role	Key Functional Implementation
HTML page structure	Storefront Root	Hero section, navigation bar (Home, Our Cakes, About, Baking Classes, Contact), welcome and brand-note sections, footer
Cake collection section	Catalog	Four category cards (Birthday & theme, Chocolate, Anniversary & celebration, Custom creations) linked to the enquiry form
Enquiry form section	Enquiry Engine	Name, contact, category, flavour, weight, event date, theme, image URL and budget fields with required-field validation
CSS stylesheet	Design System	Colour tokens, typography, responsive grid and component styling
JavaScript logic	Behaviour	Validation, smooth scrolling and URL-encoded WhatsApp message generation
Image assets	Media	Cake photographs used across the homepage, collection and classes sections

6.2 Interactive Enquiry Engine Algorithm Overview

The enquiry engine gathers the details a baker needs to quote a cake, verifies that all required fields are present, and composes a structured, URL-encoded message addressed to the business's WhatsApp number (+91 95978 33830). The "Not decided yet", "Not sure yet" and "Prefer to discuss" choices let customers who are still deciding proceed without friction.

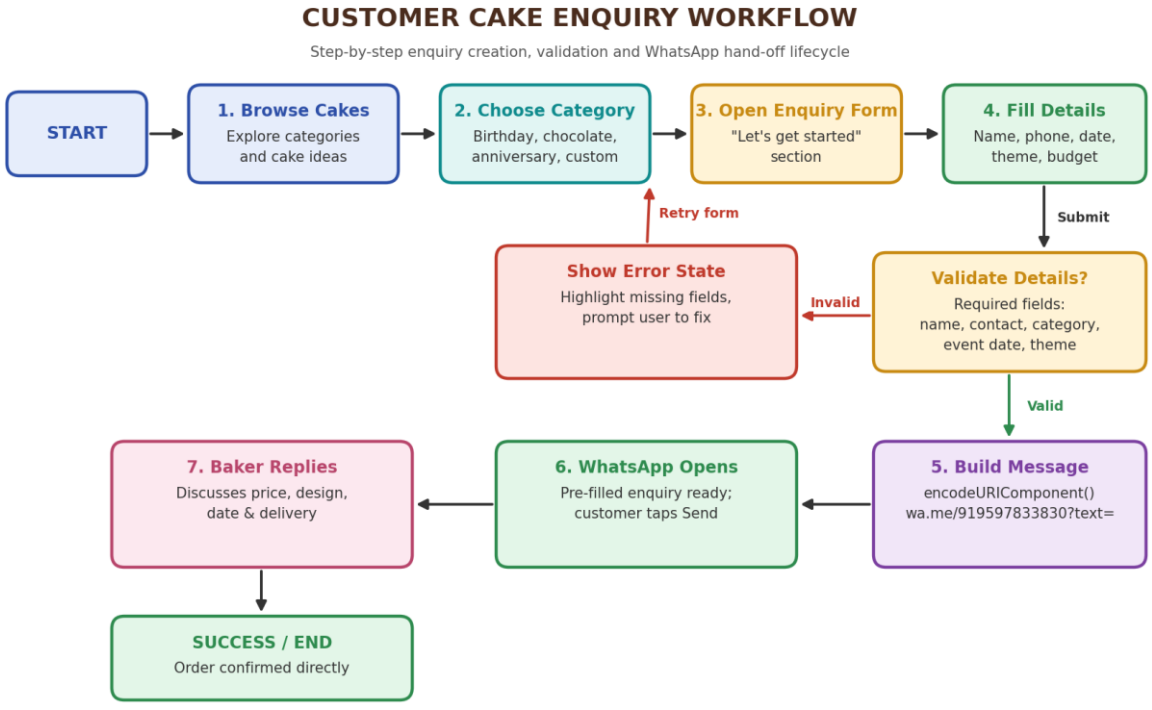


Figure 6.1: Flowchart of Customer Cake Enquiry Workflow

Table 6.2: Enquiry Engine Rule Matrix & Validation Criteria

Validation Dimension	Rule Applied	Error Mitigation Action
Customer Name	Required; must not be empty	Prompts the visitor to enter a name
Contact Number	Required; must not be empty	Prompts for a reachable number
Cake Category	Required; one of the listed styles	Prevents continuing until a category is chosen
Event Date	Required; date field	Prompts for the celebration date
Theme / Design Idea	Required; must not be empty	Prompts for a short description of the idea
Optional Fields	Flavour, weight, image URL, budget	Defaults to "Not decided yet" / "Prefer to discuss"

6.3 Baker Order-Handling Workflow

Since the website is enquiry-based, the owner-side workflow begins when the pre-filled WhatsApp message arrives. The baker reviews the details, shares a quotation, confirms the order and arranges delivery, all directly with the customer and outside the website. Content updates, such as new photographs or text, are made in the Git repository and published automatically.

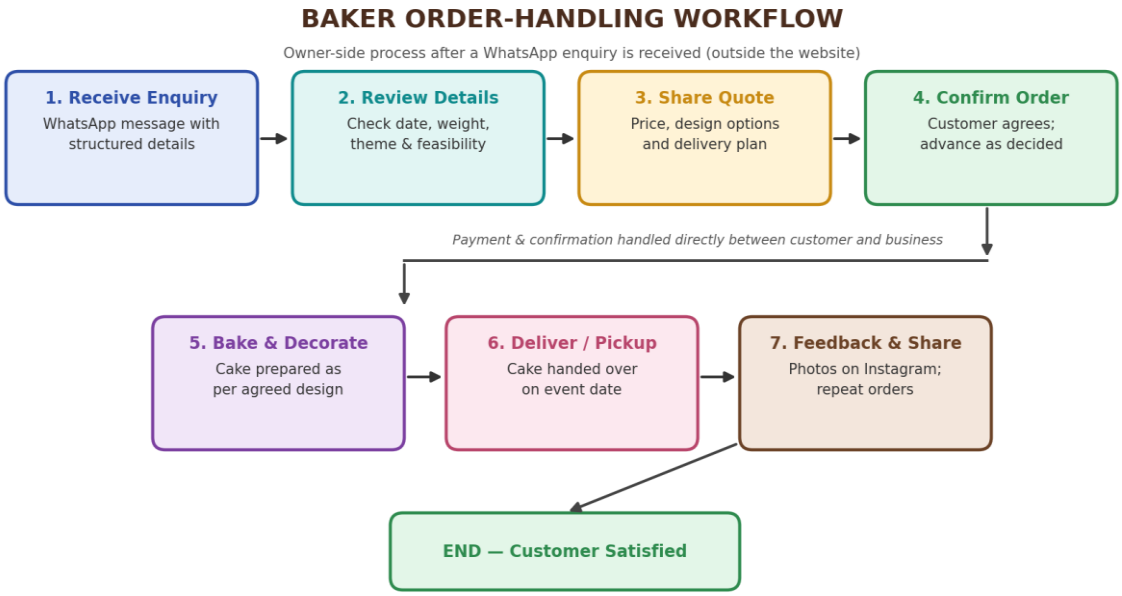


Figure 6.2: Flowchart of Baker Order-Handling Workflow

6.4 Implementation Status & Engineering Challenges Overcome

Table 6.3: Module Implementation Status & Engineering Challenges Overcome

Module	Implementation Challenge Encountered	Technical Resolution Applied
WhatsApp Message	Special characters, spaces and line breaks in form text broke the wa.me URL	Applied strict encodeURIComponent() formatting to every parameter
Enquiry Form	Customers undecided about flavour, weight or budget abandoned the form	Added "Not decided yet", "Not sure yet" and "Prefer to discuss" choices
Responsive Layout	Cake imagery and multi-column sections overflowed on small screens	Implemented fluid grids and breakpoints for tablet and mobile
Imagery	Final business photographs were not yet available	Used royalty-free sample photography as visual inspiration, ready to be replaced with approved photos

7. VERIFICATION, QUALITY ASSURANCE & TESTING SUITE

7.1 Testing Strategy & Quality Standards

Quality assurance followed a systematic multi-tier protocol: functional verification of the enquiry workflow, link and navigation checks, cross-browser and cross-device compatibility audits, and a structured review of accessibility, search metadata and security of the delivered pages.

7.2 Comprehensive Functional Test Execution Matrix

Table 7.1: Comprehensive Functional Test Execution Matrix

Test ID	Feature Tested	Input / Action	Expected Result	Status
TC-01	Navbar & Smooth Scroll	Click Our Cakes, About, Baking Classes, Contact links	Smooth viewport transition to the chosen section	Passed
TC-02	Cake Collection	Browse the four category cards	Cards render with image, title and description	Passed
TC-03	Form Validation	Submit with required fields empty	Required fields are flagged and submission is blocked	Passed
TC-04	WhatsApp Message	Submit a valid enquiry	WhatsApp opens with the entered details pre-filled	Passed
TC-05	WhatsApp CTAs	Tap navbar button and floating icon	Chat with +91 95978 33830 opens	Passed
TC-06	Social Links	Tap Instagram and YouTube links	Correct channels open in a new view	Passed
TC-07	Baking Classes	Tap an "Enquire about classes" link	WhatsApp opens with a pre-filled class enquiry	Passed

7.3 Cross-Browser & Device Responsiveness Matrix

Table 7.2: Cross-Browser & Device Responsiveness Compatibility Matrix

Platform / Browser	Viewport / Device	Observed Rendering Fidelity	Verification Result
Google Chrome (Desktop)	1920x1080 (Desktop)	Clean layout, smooth scrolling and transitions	Fully Verified
Mozilla Firefox (Desktop)	1440x900 (Laptop)	Consistent grid alignment and colour rendering	Fully Verified
Microsoft Edge (Desktop)	1366x768 (Laptop)	Identical rendering to Chrome	Fully Verified
Mobile Safari (iOS)	390x844 (iPhone)	Readable text, tappable buttons, WhatsApp opens	Fully Verified
Android Chrome	412x915 (Android phone)	Responsive layout; WhatsApp deep-link launches app	Fully Verified

7.4 Quality Audit, Accessibility & Performance Baseline

Table 7.3: Quality Audit & Technical Compliance Baseline

Audit Dimension	Measured Feature	Technical Compliance Status
Performance	Lightweight static pages served from the GitHub Pages CDN	No server round-trips or database queries on page load
Accessibility	Skip-to-content link, semantic headings, mobile-friendly tap targets	Aligned with WCAG 2.1 principles
Best Practices	HTTPS enforced by GitHub Pages; no stored personal data	No mixed content or credential handling
SEO	Page title, meta description, Open Graph tags, theme colour and viewport meta tag	Optimised for local Hosur cake search and link sharing

8. DEPLOYMENT, WEB OPERATIONS & LOW-COST INFRASTRUCTURE

8.1 Continuous Deployment Pipeline on GitHub Pages

The production website is continuously deployed through GitHub Pages. Every push to the main branch of the repository triggers an automated publishing process that packages the static files, applies HTTPS and refreshes the global content delivery network.

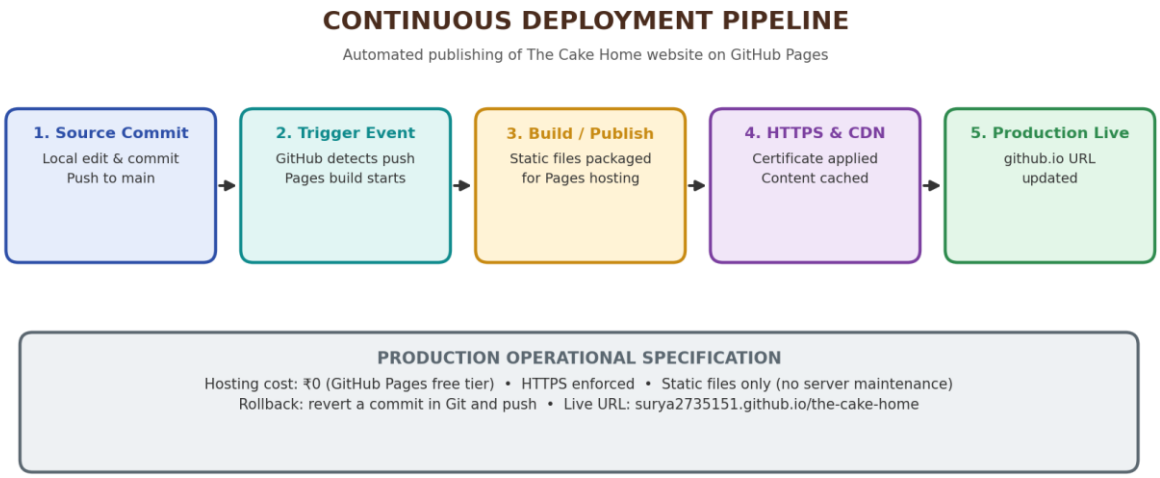


Figure 8.1: Build and Continuous Deployment Pipeline on GitHub Pages

8.2 Production Deployment Parameters Matrix

Table 8.1: Production Deployment Parameters & Environment Matrix

Deployment Parameter	Production Operational Specification
Live Production URL	https://surya2735151.github.io/the-cake-home/
Source Code Repository	https://surya2735151.github.io/the-cake-home/ (GitHub project: the-cake-home)
Hosting Infrastructure	GitHub Pages static hosting with global CDN
Application Runtime	None required (pure client-side HTML, CSS and JavaScript)
Database	None by design; no customer data is stored
SSL / TLS Certificate	Automatic HTTPS provided and renewed by GitHub Pages
Monthly Hosting Expenditure	₹0.00 (Zero recurring operational cost under free-tier allowances)
System Availability	No downtime observed throughout development, UAT and handover

9. CLIENT INTERACTION, MEETING RECORDS & CHANGE MANAGEMENT

9.1 Client Meeting and Progress Report (Mandatory Syllabus Format Table 3)

Table 9.1: Client Meeting & Progress Report (Mandatory Syllabus Format Table 3)

Meeting / Event	Date	Discussion / Work Completed	Client Feedback & Actions
Initial Briefing	12 Aug 2026	Understood how cake orders arrive through WhatsApp and the need for structured enquiries	Client asked for a warm, premium look and direct WhatsApp access
Design Review	28 Aug 2026	Presented layout mockups, colour palette and typography	Approved cocoa-and-rose palette; requested a Baking Classes section
Core Build Demo	18 Sep 2026	Demonstrated homepage, cake collection and enquiry form	Praised the organised form; suggested optional flavour and budget fields

Meeting / Event	Date	Discussion / Work Completed	Client Feedback & Actions
Integration Review	07 Oct 2026	Showcased WhatsApp message generation and social links	Validated pre-filled message on mobile; requested a floating WhatsApp icon
Final Sign-Off	29 Oct 2026	Live production walkthrough and UAT sign-off	Client expressed satisfaction; fee settled on 30 Oct 2026

9.2 Formal Client Change Request Log

Table 9.2: Formal Client Change Request (CR) Log

CR #	Date Requested	Modification	Technical Resolution Applied	Status
CR-01	29 Aug 2026	Add a Baking Classes section	Created classes section with three workshop ideas and per-class WhatsApp links	Approved
CR-02	20 Sep 2026	Make flavour, weight and budget optional with "not sure" choices	Extended the form schema and message template with optional fields	Approved
CR-03	08 Oct 2026	Add a floating WhatsApp icon on every screen	Implemented a fixed-position WhatsApp button linked to wa.me	Approved

10. REVENUE GENERATION MODEL & PAYMENT DETAILS

A revenue generation model describes how a project creates financial value, who pays for it, and how the money received is accounted for. This chapter distinguishes two flows of value: **development service revenue**, the professional fee paid to the developer (recorded in Tables 10.1 to 10.8), and **potential business revenue**, the cake orders The Cake Home may obtain through the website, which belongs entirely to the client and is not counted as project income. No sales figures or order volumes are claimed because no such records were supplied.

10.1 Revenue Breakdown (Mandatory Syllabus Format Table 2)

Table 10.1: Revenue Breakdown (Mandatory Syllabus Format Table 2)

Particulars (Mandatory Format Table 2)	Amount (INR)
Total Quoted Project Value (Agreed Remuneration)	₹1,000.00
Advance Payment Received (single-instalment settlement)	₹0.00
Final Settlement Received on 30 October 2026 (UPI)	₹1,000.00
Total Revenue Realized & Verified	**₹1,000.00**
Pending Amount (Agreed Value − Total Received)	₹0.00
Expenditure on Frameworks, Hosting & Licenses	₹0.00 (Open Source & Free Tier)
Less: Charitable Contribution to KPR Charities (02 Nov 2026)	₹1,000.00
Remaining Project Earnings after Contribution	₹1,000.00

Particulars (Mandatory Format Table 2)	Amount (INR)
Payment Mode & Settlement Status	Electronic Transfer (UPI), 100% Fully Settled

10.2 Labor Effort & Billable Valuation Matrix

Table 10.2: Engineering Labor Effort & Commercial Valuation Matrix

Team Member	Role & Primary Technical Responsibilities	Effort (Hrs)	Pro-rata Value
BALA GANESH M (24AM007)	Requirement analysis, UI/UX design, front-end development, enquiry form & WhatsApp integration, QA testing, GitHub Pages deployment	45 Hours	₹1,000.00
TOTAL	**Cumulative Engineering Project Investment**	**45 Hours**	**₹1,000.00**

10.3 Website Development Services and Revenue Breakdown

The development fee was organised into nine service components, each corresponding to a distinct stage of the work. The sum of all nine items equals the total agreed project value.

Table 10.3: Website Development Services and Revenue Breakdown

S. No.	Development Service	Description of Work	Amount (₹)
1	Requirement Analysis and Project Planning	Discussion with the client, study of the business, content, social links and contact details, scope definition	₹150
2	UI/UX Design and Website Layout	Colour palette, typography, page structure and visual hierarchy for a premium cake brand	₹125
3	Homepage and Navigation Development	Hero, welcome and brand-note sections, navigation bar and calls to action	₹125
4	Cake Gallery and Category Integration	Organised presentation of cake photographs and categories	₹125
5	Customized Cake Enquiry Form	Form fields for name, contact, category, flavour, weight, date, theme, image and budget with validation	₹125
6	WhatsApp Integration	Pre-filled WhatsApp message, "Order on WhatsApp" button and floating icon	₹100
7	Responsive Website Development	Layouts adapted for desktop, tablet and mobile screens	₹100
8	Testing and Quality Assurance	Verification of links, form behaviour, WhatsApp redirection and device display	₹75
9	Website Deployment and Final Handover	Publication on GitHub Pages and handover to the client	₹75
	Total Agreed Project Value (Sum of Items 1–9)		**₹1,000**

10.4 Total Project Cost, Payment Details & Expenses

Table 10.4: Total Project Cost and Payment Details

Particulars	Amount (₹)
Total Agreed Project Value	₹1,000
Advance Payment Received (Date: not applicable; Mode: UPI)	₹0
Final Payment Received (Date: 30 October 2026; Mode: UPI)	₹1,000
Total Amount Received (Advance + Final)	**₹1,000**
Pending Amount (Agreed Value − Total Received)	₹0

The fee was paid in a single instalment on **30 October 2026**, so the advance row is recorded as ₹0 and the full amount appears as the final payment. Only verified, paid costs are treated as expenses; free tools, free hosting and services already available to the developer are not classified as paid expenses.

10.5 Net Project Revenue Calculation

Net project revenue is calculated from the money actually received, not from the agreed value: (i) Total Project Revenue Received = ₹1,000; (ii) Total Project Expenses = ₹0; (iii) Net Project Revenue = (i) − (ii) = ₹1,000.

Table 10.6: Net Project Revenue Calculation

Particulars	Amount (₹)
Total Project Revenue Received	₹1,000
Less: Total Project Expenses	₹0
Net Project Revenue	**₹1,000**

10.6 Charitable Contribution – KPR Charities

As part of this project, a portion of the project earnings has been allocated to **KPR Charities**, reflecting the view that earnings from professional work can be shared with the wider community. The contribution is treated separately from development expenses: expenses are costs required to produce the website, whereas the contribution is a voluntary allocation from the earnings that remain after those costs. It is therefore deducted only once and is not a website development expense.

Table 10.7: Charitable Contribution Details

Particulars	Details
Charity Organization	KPR Charities (shown on the receipt as "K P R CHARITIES")
Contribution Amount	₹1,000 (Rupees One Thousand only)
Contribution Date	02 November 2026, 8:56 a.m.
Payment Method	UPI (Unified Payments Interface) through Google Pay
UPI Transaction ID	630640535062
Google Transaction ID	CICAgPi5pKeSCg
Remitter (as shown on the receipt)	Mr Akilan A, Indian Bank account ending 4052
Payment Status	Completed – the receiver's bank confirmed deposit of the amount to K P R CHARITIES's bank account

Charitable giving supports community welfare in ways individual projects cannot achieve alone, and it reflects an important attitude for future engineers: technical outcomes are means of creating positive impact. Net Project Revenue (₹2,000) less the Charitable Contribution (₹1,000) leaves Remaining Project Earnings of **₹1,000**.

Table 10.8: Revenue Distribution Summary

S. No.	Financial Particulars	Amount (₹)
1	Total Agreed Project Value	₹1,000
2	Total Amount Received (30 October 2026)	₹1,000
3	Total Development Expenses	₹0
4	Contribution to KPR Charities (02 November 2026)	₹1,000
5	**Remaining Project Earnings (Row 4 − Row 5)**	**₹1,000**

10.7 Revenue Opportunities and Business Value for the Client

The website creates several **potential** opportunities for The Cake Home. These are expected benefits based on the website's features, not proven financial results, because the conversion of an enquiry into an order depends on the business's own pricing, availability and service.

Table 10.9: Summary of Revenue Opportunities Created by the Website

Opportunity	What the Feature Does	Potential Benefit to The Cake Home
Online visibility	Provides a professional, shareable web presence	Greater awareness and credibility among prospective customers
Customized cake enquiries	Collects structured cake requirements	Faster, better-informed quotations and responses
Cake category promotion	Presents cake styles and occasions	Wider range of occasions for which customers consider the business

Opportunity	What the Feature Does	Potential Benefit to The Cake Home
WhatsApp communication	Opens a direct conversation with the business	Personal discussion that can lead to order confirmation
Social media integration	Connects the website with Instagram and YouTube	Continuous visual marketing linked to an enquiry channel
Customer retention	Offers a permanent place to return and refer others	Potential repeat orders and referrals

The website does not confirm orders or accept payments; those remain directly between the customer and the business. Future enhancements such as an online advance-payment option, order tracking, a custom domain or baking class booking would be subject to the client's needs and budget.

11. CLIENT VERIFICATION & AUTHENTICITY AUDIT

11.1 Business Verification Evidence

To establish authenticity, all factual details regarding The Cake Home were gathered through direct consultations with the proprietor. The client reviewed the production website and issued formal approval.

Table 11.1: Client Business Verification & Authenticity Audit Matrix

Evidence Dimension	Authenticity Verification Details	Verification Status
Business Establishment	The Cake Home cake business, Hosur, Tamil Nadu	Verified through client consultations
Client Contact	Proprietor Mr. Akilan A (+91 95978 33830, WhatsApp)	Verified via Telephony / WhatsApp
Production URL	Active deployment at https://surya2735151.github.io/the-cake-home/	Publicly Accessible
Social Channels	Instagram: @the__cake_home; YouTube: @thecakehome	Linked from website
Commercial Settlement	₹1,000 UPI payment on 30 October 2026, record documented in Annexure C	Fully Confirmed

11.2 GST Status & Micro-Enterprise Regulatory Compliance

The Cake Home is a small, home-scale business, and the commercial billing of ₹1,000 represents an educational freelance software service rendered by a student developer whose receipts fall below the mandatory threshold for Goods and Services Tax (GST) registration. No GST has been charged on the invoice.

12. PROJECT DELIVERY, TRAINING & ACCEPTANCE

12.1 Deliverables Handover Inventory

Table 12.1: Formal Deliverables Handover Inventory Matrix

Item No	Handover Asset Description	Technical Format	Handover Status
D-01	Production Website	Responsive static site deployed on GitHub Pages	Delivered Live
D-02	Source Code Repository	Git repository with commit history	Delivered to Client
D-03	Enquiry Form & WhatsApp Bridge	JavaScript form validation and wa.me message builder	Configured & Active
D-04	Social Media Integration	Instagram and YouTube links in contact section and footer	Integrated in Website
D-05	Content Update Guide	Short instructions for replacing text and photographs	Handed Over

12.2 Client Training Protocol & Acceptance Criteria

Table 12.2: Client Demonstration & Acceptance Protocol

Demonstration Stage	Training Activity Conducted	Client Acceptance Sign-Off
Stage 1: Public Website	Walkthrough of homepage, cake collection and navigation	Client confirmed high visual appeal
Stage 2: Enquiry Form	Simulated cake enquiry with required and optional fields	Client confirmed completeness of requirements captured
Stage 3: WhatsApp Bridge	Verified pre-filled WhatsApp message receipt on the proprietor's phone	Client confirmed fast and clear enquiry delivery
Stage 4: Content Updates	Explained how to update text and photographs through the repository	Client understood the update process

13. OUTCOMES, LEARNINGS, FUTURE SCOPE & CONCLUSION

13.1 Student Learning Outcomes

- **Front-End Mastery:** Gained in-depth expertise in semantic HTML5, responsive CSS3 layouts and vanilla JavaScript for validation and dynamic message generation.
- **Integration Skills:** Implemented WhatsApp Click-to-Chat deep links with correct URL encoding and connected social media channels to a business website.
- **Client-Consultative Engineering:** Acquired real-world experience translating a small business's everyday workflow into formal requirements and an agreed scope.
- **Zero-Cost Deployment:** Built a free, HTTPS-enabled deployment pipeline on GitHub Pages with Git version control.

13.2 Future Scope & Enhancement Roadmap

- **Online Advance Payment:** Adding a payment gateway (UPI / Razorpay) so customers can pay an advance after the price is agreed.
- **Order Tracking & Reminders:** Introducing order status updates and event-date reminders for customers.
- **Custom Domain & Real Gallery:** Registering a custom domain and replacing sample imagery with the business's own approved cake photographs.
- **Baking Class Booking:** Adding a booking feature for the baking classes and workshops presented on the website.

13.3 Project Conclusion

The Cake Home Website successfully gave a small cake business a professional, elegant and secure online presence and replaced unstructured enquiries with a complete, WhatsApp-ready request. The platform presents the cake collection clearly, connects Instagram, YouTube and WhatsApp in one destination, and requires no recurring cost. Order confirmation and payment remain directly between the customer and the business. The project met its academic and commercial objectives with full client satisfaction, full revenue realization of ₹1,000 and a completed contribution of ₹1,000 to KPR Charities.

14.1 Project Approval / Verification Sign-Off Matrix (Mandatory Syllabus Format Table 5)

This matrix records the formal institutional verification, guide endorsement and client sign-off for the completed client-based software engineering project in accordance with institutional evaluation rubrics.

Table 14.2: Project Approval / Verification Sign-Off Matrix (Mandatory Format Table 5)

Particulars (Table 5)	Student Engineering Team	Faculty Project Guide	Client Representative
Candidate / Evaluator Name:	1. BALA GANESH M (24AM007)	Dr. KARTHICK PANNEERSELVAM	Mr. Akilan A
Academic Designation:	Student Candidate Dept of CSE (AIML)	Associate Professor Dept of CSE (AIML)	Proprietor & Baker The Cake Home
Institutional Affiliation:	KPR Institute of Engg & Tech Coimbatore – 641 407	KPR Institute of Engg & Tech Coimbatore – 641 407	The Cake Home Hosur, Tamil Nadu
Official Signature:	1. _____	_____	_____
Formal Sign-off Date:	02 November 2026	02 November 2026	02 November 2026

Table 14.3: Faculty Guide Verification Box

Faculty Guide Signature	Date & Institutional Seal
_____ Dr. KARTHICK PANNEERSELVAM Associate Professor, Dept of CSE (AIML)	Date: 02 November 2026 KPR Institute of Engineering and Technology Coimbatore – 641 407

15. COMPREHENSIVE ANNEXURES & ACADEMIC REFERENCES

Annexure A: Client Verification & Studio Profile

The Cake Home is an active cake business based in Hosur, Tamil Nadu, offering customized cakes, birthday cakes, anniversary cakes, celebration cakes and theme-based cakes, along with baking classes. Orders are discussed and confirmed through WhatsApp. The website was developed to give the business a professional online presence and a structured enquiry channel, and is publicly accessible at https://surya2735151.github.io/the-cake-home/.

Annexure B: Visual Interface Walkthrough & Project Screenshots

This annexure documents the verified visual interfaces of the deployed The Cake Home website. All screenshots represent authentic interfaces captured from the production build.

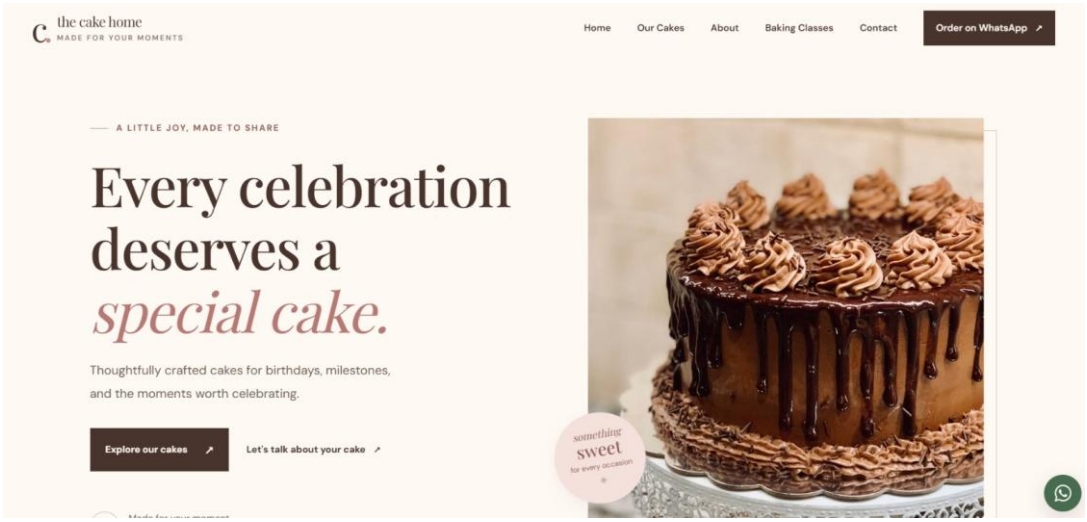


Figure B.1: The Cake Home Website Homepage & Hero Section

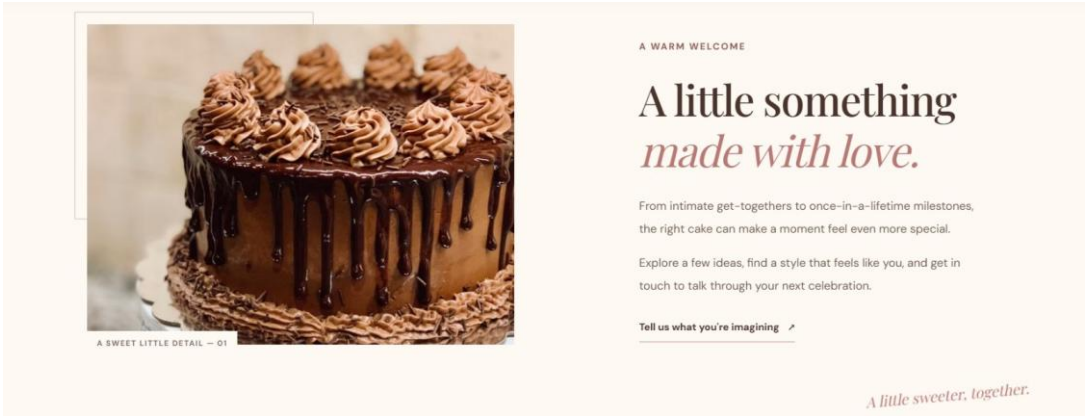


Figure B.2: "A Warm Welcome" Section Inviting Visitors to Explore Cake Ideas

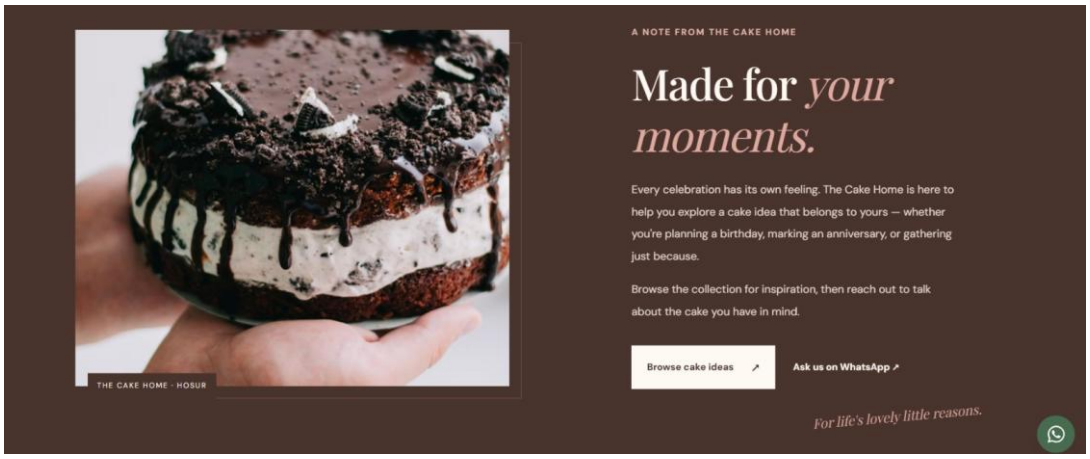


Figure B.3: "A Note from The Cake Home" Section with Browse and WhatsApp Links

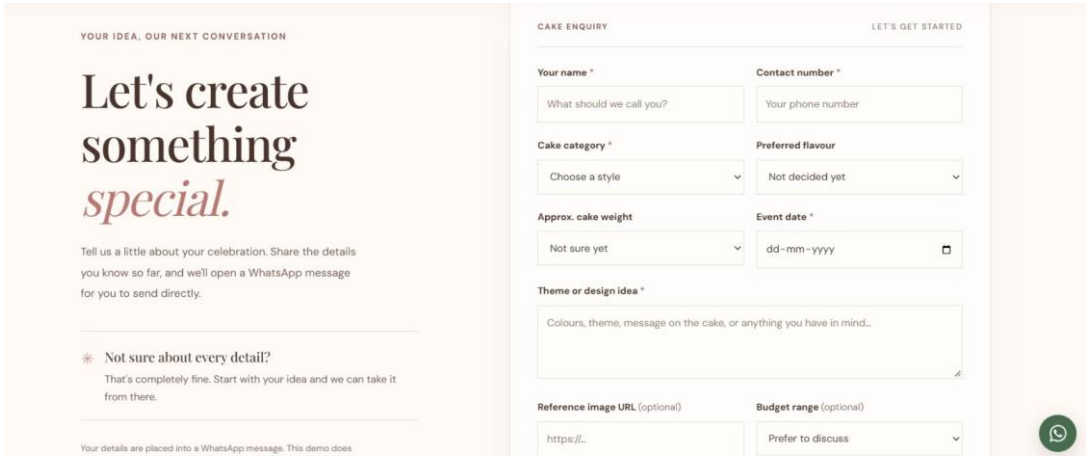


Figure B.4: Customized Cake Enquiry Form with Pre-filled WhatsApp Hand-off

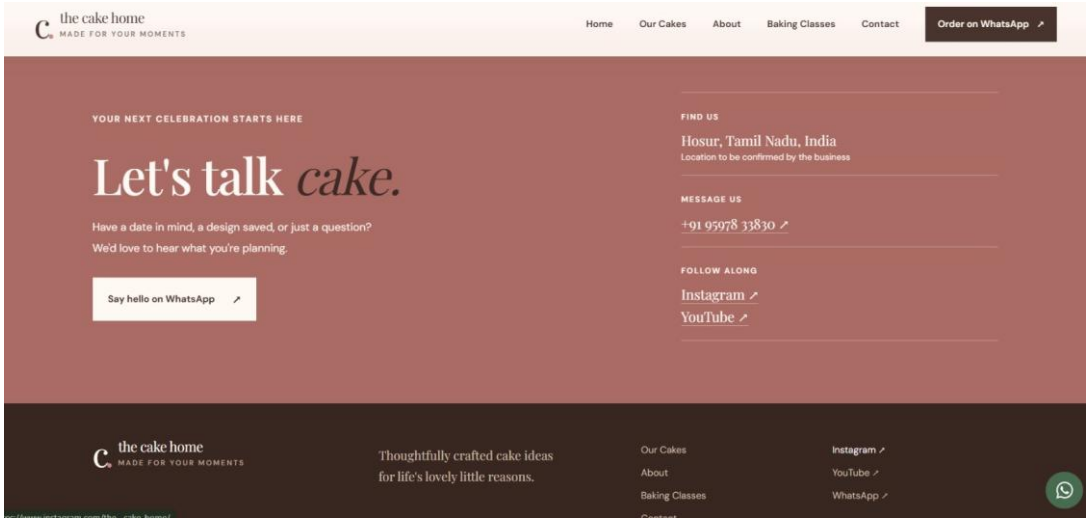


Figure B.5: WhatsApp Communication, Location and Social Media Links

Annexure C: Invoice Details, Commercial Quotation, Testimonial & Client's Appreciation

INVOICE

Invoice No : INV-2026-001
Date : 30 October 2026

Bill To
MR. AKILAN A
Proprietor & Baker
The Cake Home
Hosur, Tamil Nadu, India
Phone: +91 95978 33830

From
BALA GANESH M
Register No: 24AM007
B.E. Computer Science and Engineering (Artificial Intelligence and Machine Learning)
KPR Institute of Engineering and Technology, Coimbatore – 641 407, Tamil Nadu, India

Project Title	THE CAKE HOME – CAKE BUSINESS WEBSITE (Web Application Design and Development)
Client	The Cake Home, Hosur (Proprietor: Mr. Akilan A)
Project Period	10 August 2026 – 29 October 2026
Payment Terms	As agreed (₹1,000, all-inclusive)

Sl. No.	Description	Details	Amount (₹)
1	Requirement Analysis and Project Planning	Discussions, content, scope	150
2	UI/UX Design and Website Layout	Palette, typography, layout	125
3	Homepage and Navigation Development	Hero, sections, navbar, CTAs	125
4	Cake Gallery and Category Integration	Four cake categories	125
5	Customized Cake Enquiry Form	Nine fields with validation	125
6	WhatsApp Integration	Pre-filled message, icon	100
7	Responsive Website Development	Desktop, tablet, mobile	100
8	Testing and Quality Assurance	Links, form, WhatsApp, devices	75
9	Website Deployment and Final Handover	GitHub Pages and handover	75
	Domain and Hosting	Free GitHub Pages subdomain	0
	Total Agreed Project Value		**₹ 1,000**

Amount in Words: Rupees One Thousand Only

Issued By,	Client Acknowledgement,	Date: 30 October 2026
BALA GANESH M Register No: 24AM007 Project Developer	**MR. AKILAN A** Proprietor, The Cake Home Signature	Official Business Seal / Stamp:

Payment Proof

The payment record below documents the electronic settlement of the agreed development fee. The completed charitable contribution to KPR Charities is separately recorded in Table 10.7.

Figure C.1: Client Electronic Payment Record (₹2,000)

Payment Particulars	Recorded Details
Amount Received	₹1,000 (Rupees Two Thousand only)
Date of Receipt	30 October 2026
Payment Mode	UPI (Unified Payments Interface), electronic transfer
Paid By	Mr. Akilan A, Proprietor, The Cake Home, Hosur
Paid To	BALA GANESH M (24AM007), Project Developer
Against	Invoice INV-2026-001 dated 30 October 2026
Settlement Status	Credited in full – no pending balance

Table C.1 itemises the commercial billing and payment status. Payment has been received in full as of the date of this report.

PAYMENT PROOF

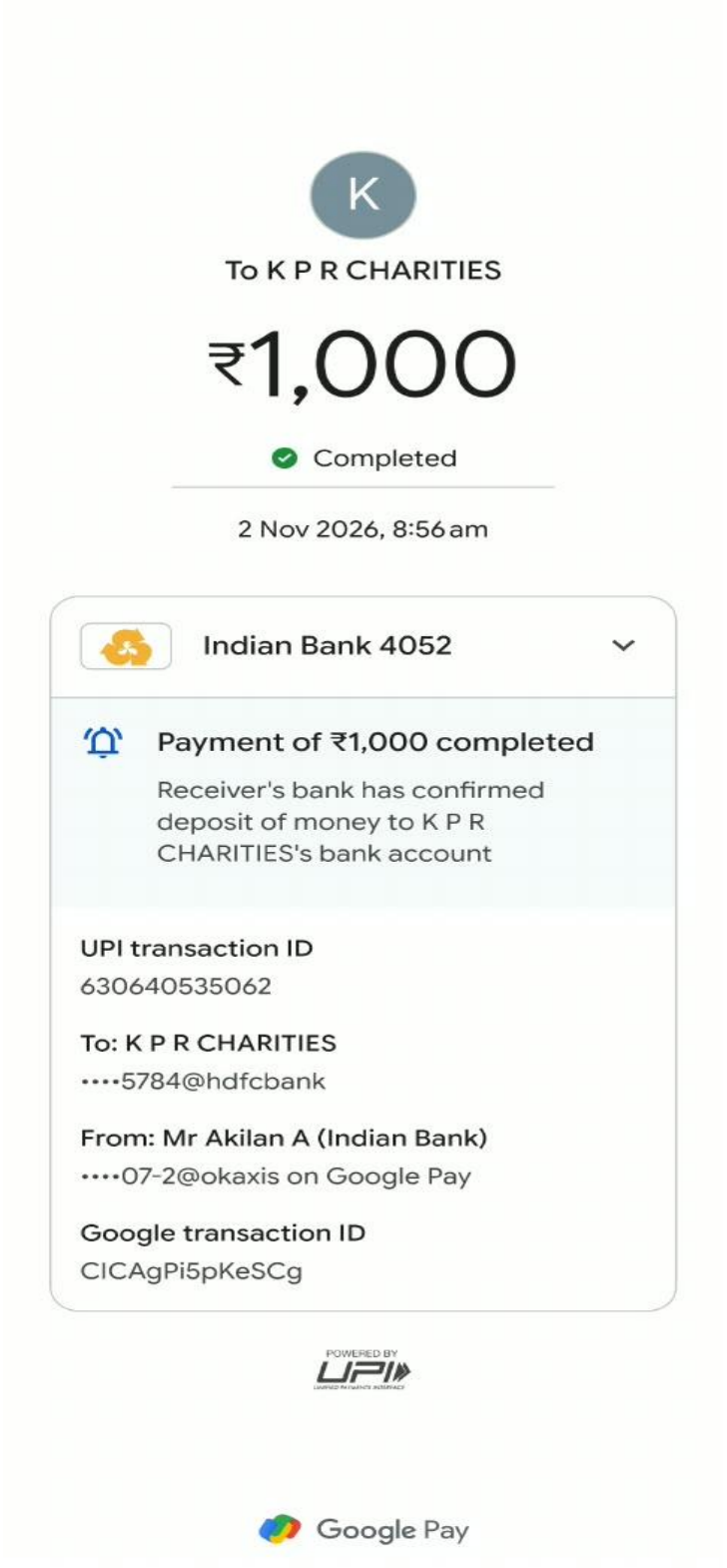


Table C.1: Itemised Billing & Settlement Status

Item	Service Description	Amount	Settlement Status
1	Requirement Analysis and Project Planning	₹150	Completed & Settled
2	UI/UX Design and Website Layout	₹125	Completed & Settled
3	Homepage and Navigation Development	₹125	Completed & Settled
4	Cake Gallery and Category Integration	₹125	Completed & Settled
5	Customized Cake Enquiry Form	₹125	Completed & Settled
6	WhatsApp Integration	₹100	Completed & Settled
7	Responsive Website Development	₹100	Completed & Settled
8	Testing and Quality Assurance	₹75	Completed & Settled
9	Website Deployment and Final Handover	₹75	Completed & Settled
10	Domain & Hosting (GitHub Pages)	₹0	Completed (Free Tier)
TOTAL	**Total Agreed Project Value Realized**	**₹1,000**	**100% Fully Settled**

Client Testimonial

"Bala Ganesh built our cake website after understanding how we take orders on WhatsApp. The enquiry form sends us the customer's details in one organised message, and the website shows our cakes and links to our Instagram and YouTube pages. We are happy with the work."

— **Mr. Akilan A**, Proprietor, The Cake Home, Hosur

Table C.2: Testimonial Verification

Testimonial Verification	Details
Client Name	Mr. Akilan A
Designation	Proprietor & Baker, The Cake Home, Hosur
Date of Statement	30 October 2026
Client Signature	_____

Annexure D: References

[1] GitHub Inc., 'GitHub Pages Documentation: Hosting a Static Site,' https://docs.github.com/pages, 2026.

[2] Mozilla Developer Network, 'HTML, CSS and JavaScript Reference,' https://developer.mozilla.org, 2026.

[3] Meta Platforms Inc., 'WhatsApp Click to Chat (wa.me links),' https://faq.whatsapp.com/5913398998672934, 2026.

[4] World Wide Web Consortium, 'Web Content Accessibility Guidelines (WCAG) 2.1,' https://www.w3.org/TR/WCAG21/, 2026.

[5] Mozilla Developer Network, 'encodeURIComponent() – JavaScript Reference,' https://developer.mozilla.org/docs/Web/JavaScript/Reference/Global_Objects/encodeURIComponent, 2026.

[6] Project-specific requirements, client discussions, configuration and implementation records documented in the source report, Hosur, Tamil Nadu, 2026.